

Yonghyeon Choi

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EDUCATION

University of California, San Diego

B.S. in Mathematics, Computer Science | GPA: 3.888/4.000

La Jolla, CA

Expected Graduation: March 2027

EXPERIENCE

Software Engineer Intern

Samsung SDS

Jun. 2026 - Jul. 2026

Seoul, South Korea

- Built a full-stack web application with an **HTML/CSS/JavaScript** frontend and a **JavaScript/Node.js** backend, giving employees a simple interface to access and interact with an AI agent for internal business questions
- Grounded the AI agent in internal business knowledge by preparing **400+** historical Q&A records for **RAG** retrieval and configuring domain specific instructions for consistent responses across various business scenarios
- Implemented the request/response workflow through **REST APIs**, authenticating users against an internal system, routing queries to the AI agent, and delivering responses to users by enterprise email service and web interface

Software Engineer Intern, Validation & Reliability

Solidigm

Jul. 2025 - Sep. 2025

Rancho Cordova, CA

- Rebuilt and migrated a legacy Python test automation plugin to the team's repository, reducing it from **~3,000 to ~300 lines** by analyzing its role in the overall test workflow and retaining only the commands required for its core functionality
- Integrated the rebuilt plugin as a new node in the existing messaging architecture, conforming to its **queue-based** messaging design to enable communication between the backend and remote SSD test environment
- Automated remote test setup and execution in Python across **Linux** and **Windows**, using **Git** and system commands to prepare environments, retrieve requested software revisions, launch tests, and log execution status

PROJECTS

Onyeon | JavaScript, REST APIs

[Website](#) / [GitHub](#)

- Built a full-stack web application for a curated dating service that surfaces compatible matches based on user preferences
- Recruited **100 professionals** across New York, San Francisco, and Los Angeles within **three weeks** through targeted outreach, then conducted **1:1 Zoom interviews** to understand preferences and refine matching criteria
- Developed the application's client and server architecture in **JavaScript**, using an **Airtable** data layer for profile retrieval, authentication, and match status updates

Geospatial Rental Market Analysis | Python, GeoPandas, OSMnx, Statsmodels

[GitHub](#)

- Created a Python geospatial data pipeline combining SanGIS, RentCast API, and OpenStreetMap data, aggregating **4,900+** rental listings into **133** sampled locations with distances to nearby amenities
- Evaluated regression models across feature combinations using AIC/BIC and calibrated a **95%** conformal prediction interval
- Applied the selected model to compare UCSD housing prices with the surrounding market under the same pricing framework

SSD Validation Configuration Optimizer | Python, PyGAD, NumPy, Pandas

Solidigm Project

- Created a Python optimization tool that parsed SSD validation JSON files into structured objects and generated test configurations with a **genetic algorithm** to maximize validation coverage
- Customized fitness scoring, selection, crossover, and mutation logic, and designed an interactive GUI to configure optimization runs and visualize coverage progression across generations

SKILLS

Languages: Python, Java, C, C++, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, FastAPI, NumPy, Pandas, GeoPandas, OSMnx, Statsmodels

Tools: Git, Linux, Docker, Kubernetes

Databases: MySQL, PostgreSQL, MongoDB, Airtable

Cloud: AWS, Vercel